MINISTRY OF EDUCATION, SINGAPORE
in collaboration with
CAMBRIDGE ASSESSMENT INTERNATIONAL EDUCATION
General Certificate of Education Advanced Level
Higher 2

ECONOMICS

9570/01

Paper 1

October/November 2023

2 hours 30 minutes

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper ask the invigilator for a continuation booklet.

Answer **all** questions.

The number of marks is given in brackets [ ] at the end of each question or part question.

Extract 2: The returns to investment made in education

Table 1: Private and social returns to investment in education by per capita income and educational level

	Private return (% per year)			Social return (% per year)		
Countries	Primary education	Secondary education	Higher education	Primary education	Secondary education	Higher education
Low-income	25.4	18.7	26.8	22.1	18.1	13.2
Middle-income	24.5	17.7	20.2	17.1	12.8	11.4
High-income	28.4	13.2	12.8	15.8	10.3	9.7

The private return is the annual rate of return to an individual on the investment made in education. The social return is the annual rate of return in the form of external benefits to society on the investment made in education.

Recent research indicates that the social rates of return in the table are under-estimates of the true returns because of the omission of benefits that cannot be easily expressed in monetary terms, such as reduced mortality rates and lower crime rates. In fact, the social rate of return to investment in education could be 50% higher than the one traditionally estimated.

Source: Returns to Investment in Education: World Bank Group, April 2018

Extract 3: Education and the demand and supply of labour

On average, the difference between the earnings of a university graduate and a high school leaver who does not enter HE is significant in most countries. This difference can, in part, be explained as the increase in productivity that HE gives to individuals. The difference in earnings can also be caused by differences in supply.

A well-trained and educated workforce is likely to be in high demand by employers. Demand will be lower for a workforce that is not well-trained or educated because employers will need to spend a significant amount of time and money training that workforce.

The higher the level of education required for a particular career, the lower the supply of labour. For example, there is a lower supply of medical doctors than nurses because of the additional training required to be a medical doctor.

Source: Principles of Economics, Rice University, September 2021

(a) With reference to Figure 1, compare the potential benefits for a graduate in the US with a Bachelor's degree to a school leaver who has a High School Diploma. [2]

(b) With reference to Extract 1 and using a supply and demand diagram, explain **one** possible reason for the higher average earnings of graduates with a Professional degree compared to those with a Bachelor's degree. [3]

(c) 'Opportunity costs may make even free schooling unaffordable for some families.' (Extract 4)

Explain **one** example of an opportunity cost that might make free schooling unaffordable. [3]

(d) Explain how asymmetric information may lead to wrong choices in the market for education. [4]

(e) The government of a low-income country wishes to increase spending on education.

With reference to Table 1, discuss whether the government should concentrate this increase in spending on primary education. [8]

(f) Discuss whether equity issues are more important than market failures as a reason for the government to intervene in the market for education. [10]

[Total: 30]

Firms may choose to cut investment in the domestic economy substantially, even as interest rates fall, if they think that output and consumption growth will slow in response to an ageing population. Should firms become pessimistic, even if central banks drive interest rates below zero, the economy could have zero growth with high unemployment for many years. This may be shown in Japan's virtually flat economic growth in recent decades and Europe's failure to recover from the global financial crisis.

An ageing population will mean more old-age dependency. But it may also bring more capital per worker leading to rising productivity and wages. Whether an ageing population is good or bad, there are no simple answers.

Source: Cost of Ageing, Finance & Development, March 2017, Vol. 54, No. 1

Extract 6: Why population ageing matters: A global perspective

For the first time in history, people aged 65 and over now outnumber children under age 5 (as shown in Figure 2). In 2006, almost 500 million people were 65 and older, accounting for about 8% of the world's population.

By 2030, the world is predicted to have 1 billion older people, accounting for 13% of the total population. While proportions of older people have typically been highest in high-income countries, the most rapid increases in older populations are now occurring in low-income countries. Between 2006 and 2030, the number of older people in low-income countries was projected to increase by 140% as compared to an increase of 51% in high-income countries.

Figure 2: Young children (age < 5) and older people (age 65+) as a percentage of global population

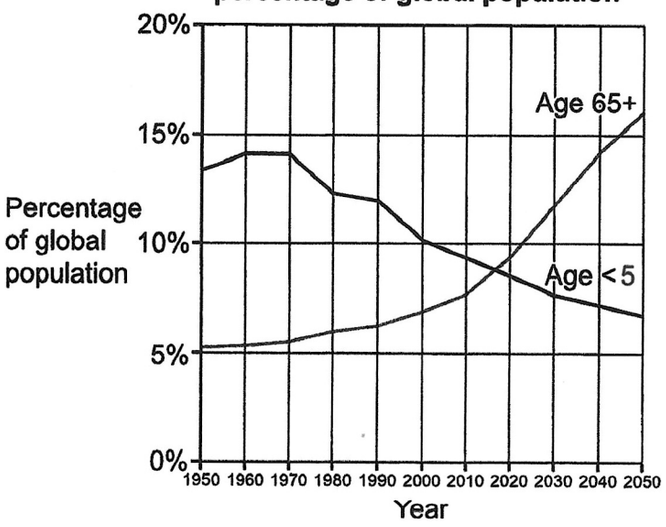
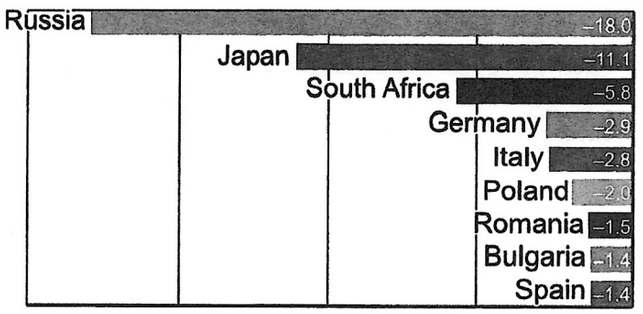


Figure 3: Projected population decline of selected countries between 2006 and 2030 (millions)



More than 20 countries were projected to experience population decline by 2030. As Figure 3 shows, for example, Russia's population was expected to shrink by 18 million between 2006 and 2030, a decrease of nearly 13%.

Source: adapted from Why Population Aging Matters: A Global Perspective, USA National Institute on Aging, March 2007

(a) With reference to Extract 5:

(i) calculate and compare the old-age dependency ratio for Singapore in 1990 with that in 2020. [3]

(ii) explain **one** reason for the change in the old-age dependency ratio for Singapore between 1990 and 2020. [2]

(b) Explain how the changes in the populations of countries shown in Figure 3 might affect aggregate supply and aggregate demand in those countries. [4]

(c) With reference to Extract 5, explain why firms may choose 'to cut investment in the domestic economy substantially, even as interest rates fall'. [3]

(d) Discuss whether the benefits to an economy of having an ageing population outweigh the costs. [8]

(e) Discuss whether immigration is the best way of promoting economic growth in an economy with an ageing population. [10]

[Total: 30]

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9570/02

Paper 2 Essays

October/November 2023

2 hours 30 minutes

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front cover of the answer booklet. If you need additional answer paper ask the invigilator for a continuation booklet.

Answer **three** questions in total, of which **one** must be from Section A, **one** from Section B and **one** from **either** Section A or Section B.

The number of marks is given in brackets [ ] at the end of each question or part question.

Section B

One or two of your three chosen questions must be from this section.

4 The exchange rate is a key policy tool in managing the economy of Singapore.

(a) Explain how an appreciation of the exchange rate might affect aggregate demand and aggregate supply in an economy. [10]

(b) Discuss whether management of the exchange rate is the most appropriate way of controlling inflation in Singapore. [15]

5 Residents in densely populated, rapidly growing cities expect annual improvements in standards of living but whether this is achievable or sustainable in the long term is uncertain.

(a) Explain how improvements in a country's material and non-material standard of living can be measured. [10]

(b) Discuss whether increases in the living standards of residents in densely populated, rapidly growing cities are achievable and sustainable in the long term. [15]

6 The use of expansionary demand-side policies designed to achieve economic growth or to lower unemployment may lead to undesirable consequences.

(a) Explain why expansionary fiscal policy designed to achieve economic growth or lower unemployment may lead to undesirable consequences. [10]

(b) Discuss whether expansionary supply-side policies would be effective in achieving the macroeconomic policy aims of an economy. [15]

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Answer three questions in total.

Section A

One or two of your three chosen questions must be from this section.

1 The amount of goods and services that can be produced in an economy is insufficient to satisfy the wants of its population. This leads to the economic consequences of scarcity and choice.

(a) Explain how a production possibility curve can be used to show the concepts of underutilisation of economic resources and opportunity cost. [10]

(b) Discuss whether it is possible to increase the total production of goods and services in an economy without resulting in environmental damage or other unintended consequences. [15]

2 Bad weather and falls in consumer incomes can have different impacts on the prices of agricultural products such as vegetables, rice and grain.

(a) Explain the different impacts on the prices of vegetables due to bad weather and falls in consumer incomes. [10]

(b) Discuss the effectiveness of different measures that might ensure stability of food prices to consumers. [15]

3 A firm's decisions and strategies are influenced by the level of competition in the industry. Its decisions and strategies might also be affected by consumers' cognitive biases and concerns about the environment.

(a) Explain why a firm considers the level of competition in the industry when making decisions about the price and output level of its product. [10]

(b) If markets fail due to a lack of competition, discuss whether consumers will be disadvantaged and what might be the most appropriate form of government intervention. [15]

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Question 1 Figure 1	© https://www.bls.gov/emp/chart-unemployment-earnings-education.htm 21 April 2021.
Question 1 Extract 1	© K Albertson; *The Economic Inefficiency of Student Fees in England: Why paying tuition fees from the public purse make financial sense*; Intergenerational Foundation; April 2017.
Question 1 Extract 2	© G Psacharopoulos, H A Patrinos; *Returns to Investment in Education: A Decennial Review of the Global Literature*; World Bank Group; April 2018.
Question 1 Extract 3	© *Principles of Economics*; Rice University; https://openstax.org/details/books/principles-economics-2e 14 October 2021.
Question 1 Extract 4	© A L Hillman, E Jenkner; *Educating Children in Poor Countries*; International Monetary Fund; https://www.imf.org/external/pubs/ft/issues/issues33/ 2004.
Question 2 Table 2	© *Population and Population Structure*; Singapore Department of Statistics, June 2023.
Question 2 Extract 5	© R Lee, A Mason; *Cost of Aging*; Finance & Development; https://www.imf.org/external/pubs/ft/fandd/2017/03/lee.htm March 2017.
Question 2 Extract 6	© P J Dobriansky et al; *Why Population Aging Matters: A Global Perspective*; National Institute on Aging; https://www.nia.nih.gov/sites/default/files/2017-06/WPAM.pdf March 2007.
Question 2 Extract 8	© R Partington; *More migrant workers needed to offset ageing population, says IMF*; The Guardian News & Media Ltd; https://www.theguardian.com/business/2018/apr/09/get-more-migrant-workers-to-offset-strain-of-ageing-population-warns-imf April 2018.

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Extract 7:

Content (These countries are most ready to deal with ageing populations; www.weforum.org) removed due to copyright restrictions.

Extract 8: More migrant workers needed to offset ageing population

An International Monetary Fund (IMF) report has stated that high-income countries such as the UK, the US and Japan risk being overwhelmed by their ageing populations. It calls on them to throw open their borders to more migrant workers in response. However, net annual migration of European Union (EU) nationals to the UK in 2017, for example, fell to its lowest level for five years. UK companies have raised growing concerns about labour shortages.

Within the next few decades, working-age adults will need to support double the number of elderly people than they do now, putting immense pressure on welfare systems and reducing potential economic output by as much as 3% by 2050.

The IMF called for more investment in education to get more people into work and to increase the chances of workers staying in a job beyond retirement age in order to increase the labour supply. Shifting cultural norms and government policies have already started this process by increasing the number of women entering and remaining in the workforce. The IMF stated, however, that more women and older workers alone may not be enough for economies to solve the problem. It said labour market participation would still eventually decline and was also likely to remain below current levels, even if more women worked and more older people stayed in work for a longer time period.

Immigration policies allow countries to control the influx of people, such as migrant workers, who want to establish residence within their borders. They include rules regarding rights of access to the country and permission to participate in the labour market.

The IMF report concluded that: 'Unless technology delivers offsetting productivity gains, these findings highlight the need for many high-income countries to rethink immigration policies to boost their labour supply, alongside policies to encourage older workers to postpone retirement.'

Source: adapted from The Guardian, 9 April 2018

Question 2: Economic impact of an ageing population

Extract 5: The costs and benefits of an ageing population

Table 2: Singapore: Population statistics (thousand)

	1990	2000	2010	2020
Total population	3047	4028	5077	5686
Resident population	2736	3273	3772	4044
0 – 19 years	851	929	918	803
20 – 64 years	1720	2109	2515	2626
65 years & over	164	235	338	614
of which 80 years & over	27	40	69	124
Old-age dependency ratio*		11	13	

Note: Figures may not sum due to rounding

*Old-age dependency ratio: Residents aged 65 years & over **per hundred** residents aged 20–64 years

Source: Singapore Department of Statistics, June 2023

An ageing population and slower labour force growth affect different economies in many ways. For example, the growth of GDP slows, working-age people pay more in tax to support the elderly, and government budgets are put under strain due to the burden of increased costs of health and retirement programmes for old people.

An ageing population puts budgetary pressure on the government and society because the support ratio, which is the ratio of workers to consumers, declines. Between 2015 and 2050 the support ratio is expected to drop by 0.26% a year in the US, 0.40% in other high-income nations, and 0.82% in China.

Public sector transfer payments in the form of pensions, health care, and long-term care pose a greater problem as populations age. These payments, even after subtracting the portion funded by tax payments from the elderly, account for a large portion of government budgets. Projections indicate that typically, these programmes will become unaffordable unless, for example, productivity is increased, taxes are raised or benefits for the elderly are reduced, such as by increasing the retirement age.

In general, the higher the proportion of their consumption that the elderly finance themselves, the less cost falls on working adults as higher taxes. Unlike the US, where people retire later and rely more on their own assets in old age, Europe has a heavy reliance on public sector transfer payments to pay for consumption by the elderly. When older people contribute little to their own consumption, either through asset income or continuing to work, it puts a heavy cost on the labour force as the population ages.

Extract 4: Educating children

Given the evident benefits of a basic education, why do so many children in so many countries not get one? According to economic theory, the under-provision of a product that is clearly desirable stems from market failure. This can arise from problems on both the demand and supply sides of the market.

The cost to households is one obvious reason why demand for education might be low, given that poor families must first meet their essential needs, such as food and shelter. Education costs could include tuition, books, school supplies and there may be expenses for transport and clothing.

Effective demand for education may be low because of the opportunity costs of educating children. Parents may need their children to work to supplement the household income, or to care for sick or dependent family members. Opportunity costs may make even free schooling unaffordable for some families.

Demand for education is also affected by its perceived value. Parents may not have enough information to assess the private return on an investment in their children's education accurately. They may see the private return as too low to justify the cost, perhaps because of the poor quality of the education available to their children. They may simply be unaware of the opportunities that exist, especially if they are poorly educated themselves.

There may be problems with supply. A government may lack sufficient resources to provide educational services or be administratively incapable of channelling resources to the schools that need them. Government-financed schools may exist in urban areas but not in rural areas or may vary greatly in quality. Such schools may be ranked as a low priority by the country's economic and political leaders. Public resources may be wrongly diverted from education to other projects that serve the interests of economic and political leaders.

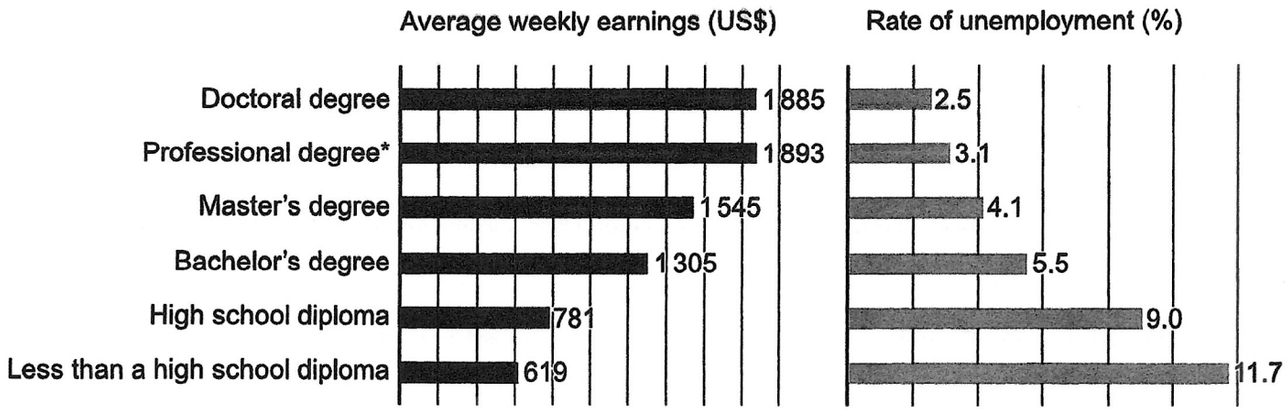
Source: Educating Children in Poor Countries, Economic Issues no 33, IMF, 2004

Question 1: Costs and benefits of education

Extract 1: The costs and benefits of Higher Education

Higher Education (HE) refers to the education of students at universities or similar establishments to graduate with a range of different degree level qualifications.

Figure 1: Earnings and unemployment rates by educational attainment, United States (US), 2020



*e.g. pharmacy, medicine, law and teaching

Sources: US Bureau of Labor Statistics and bls.gov, 21 April 2021

Costs and benefits of HE in the United Kingdom (UK)

Between 1970 and 2020, the number of students undertaking HE in the UK increased by over 400%. This increase in HE has to be paid for. Free market economists argue that, as students enjoy additional private benefits from their HE, they should cover the costs. Estimated additional career earnings of graduates compared to non-graduates, after direct taxes and student loan repayments have been paid, was over £252 000 for women and over £168 000 for men. These are the marginal private benefits.

The marginal private cost of HE, including tuition fees and other private costs of education (travel, books etc.), was over £31 000 for a student graduating in 2021. Living costs are not included, as it is argued that young people will incur these whether they go to university or not. Wages forgone must also be included as a private cost. These have been estimated at over £17 000.

It is not only the individual who benefits from HE. Society enjoys an even greater benefit from HE than does the graduate. For example, the extra tax paid by graduates compared with non-graduates is over £300 000 for women and over £260 000 for men. This is a marginal external benefit.

Source: adapted from The Economic Inefficiency of Student fees in England, Intergenerational Foundation, April 2017